

CUSTOMER CHURN BUSINESS ANALYTICS REPORT

AnalystLab Africa Data Analytics Internship

Week 1

Dataset: Telco Customer Churn Dataset

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Executive Summary

This project analysed the Telco Customer Churn dataset to understand customer behaviour and identify factors associated with customer attrition. The analysis found an overall 26.5% churn rate, with month-to-month customers and electronic-check users showing notable churn patterns. The major business concern is potential revenue and customer-value loss from continued customer attrition. Targeted retention strategies, particularly for high-risk customer segments, are recommended.

Business Understanding

Business Problem:

The company is experiencing customer churn and needs to understand which customer characteristics and services are associated with higher attrition.

Objectives:

- Identify high-churn customer segments.
- Examine the relationship between tenure, contracts, services and churn.
- Provide actionable recommendations to improve retention.

Data Overview

Item	Value
Dataset Name	Telco Customer Churn Dataset
Source	Kaggle
File Type	CSV
Number of Rows	7,043
Variables	21
Key Variables	Contract, Tenure, Internet service, Payment method, Monthly charges, Churn.

Key Business Insights

Findings	Values
Overall customer churn is significant	26.5% churn
Month-to-month customers represent the largest segment	55% of customers
Electronic-check users have the highest churn rate	45.3% churn
Fiber-optic users form a large internet-service segment	3,100 customers
Tenure matters	Average customer tenure 32.37 months

Business Risks

Risks	Values	Potential Business Impact
Customer attrition	26.5% churn	Revenue/customer lifetime value loss
Month-to-month dependency	55% of customers	Higher retention vulnerability
High-churn service segments	Significant churn contribution	Potential customer/revenue loss

BUSINESS OPPORTUNITIES

1. Convert month-to-month customers

The company has an opportunity to convert suitable month-to-month customers into longer-term contracts through loyalty incentives, discounts, improved benefits and personalized offers. Reducing the number of customers on flexible contracts could improve retention.

2. Target high-risk customers with personalized retention campaigns

Customer data can be used to identify customers who are more likely to churn and target them with proactive retention campaigns. Factors such as tenure, contract type, payment method and service usage can be combined to identify high-risk customer segments.

3. Improve the experience of high-churn customer segments

The company can investigate the reasons behind the high churn observed among electronic-check users and other high-risk service segments. Understanding whether the issue relates to payment experience, pricing, service quality or customer support could create opportunities to improve customer satisfaction and retention.

Business Recommendation

Recommendation	Action	Expected Outcome
Contract Conversion	Incentivize long-term contracts	Reduce churn
Early retention	Strengthen onboarding	Improve early retention
Payment investigation	Investigate electronic-check churn	Identify root cause
Churn monitoring	Identify high risk customers	Enable proactive intervention
Fiber-optic investigation	Analyse service experience	Reduce service-related churn
Segmented retention	Personalize interventions	Improve campaign effectiveness

Conclusion

The customer churn analysis provided valuable insights into the factors influencing customer retention. The findings highlight a significant customer churn rate and identify key customer segments that require closer attention, particularly month-to-month customers and customers using certain payment and service options. Overall, the analysis demonstrates the importance of using data-driven insights to identify churn risks, target retention efforts, and improve customer loyalty. Implementing the recommended strategies can help the business reduce customer attrition, strengthen retention, and support sustainable revenue growth.